Ions & Ionic Bonds

Exam code: 0620 & 0971

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Help me understand

The formation of ions

How are ions formed?

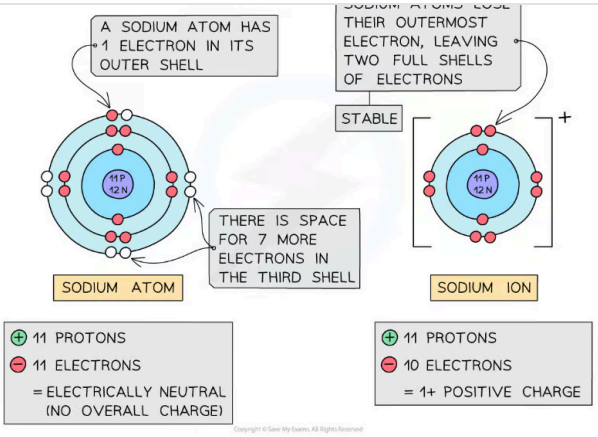
- An **ion** is an electrically charged atom or group of atoms formed by the **loss or gain of electrons**
- This loss or gain of electrons takes place to obtain a **full outer shell** of electrons
- The electronic structure of ions of elements in Groups 1, 2, 6 and 7 will be the same as that of a noble gas - such as helium, neon, and argon
- Negative ions are called **anions** and form when atoms **gain** electrons, meaning they have more electrons than protons
- Positive ions are called **cations** and form when atoms **lose** electrons, meaning they have more protons than electrons
- All metals **lose** electrons to other atoms to become positively charged ions
- All non-metals **gain** electrons from other atoms to become negatively charged ions

Formation of a cation

Stuck on something?

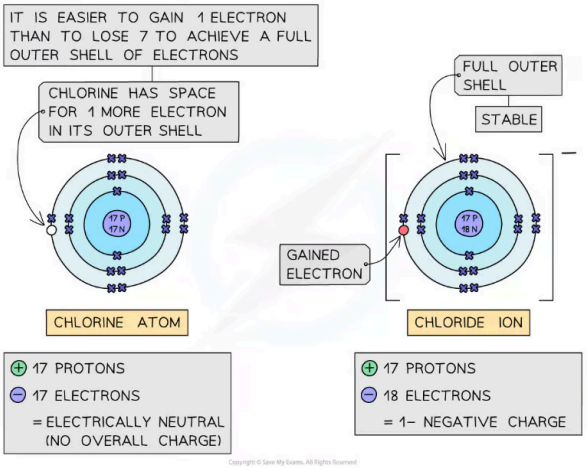
Ask questions and get help with exactly what you're revising.

Got it



Formation of positively charged sodium ion

Formation of an Anion



Formation of negatively charged chloride ion

Examiner Tips and Tricks

- We can determine the charge an ion will have from the group the element is found in:
  - Elements in Group 1 form ions with a 1+ charge
  - Elements in Group 2 form ions with a 2+ charge
  - Elements in Group 6 form ions with a 2− charge
  - Elements in Group 7 form ions with a 1− charge

Did this video help you? Yes No

What is ionic bonding?

The Formation of Ionic Bonds

- Ionic compounds are formed when metal atoms react with non-metal atoms
- Metal atoms **lose** their outer electrons which the non-metal atoms **gain** to form positive and negative ions
- The positive and negative ions are held together by strong **electrostatic forces of attraction** between **opposite** charges
- This force of attraction is known as an **ionic bond** and they hold ionic compounds together

Dot-and-cross diagrams

- **Dot and cross diagrams** are diagrams that show the arrangement of the outer-shell electrons in an **ionic** or **covalent** compound or element
- The electrons are shown as dots and crosses
- In a dot and cross diagram:
  - Often only the outer electrons are shown
  - The charge of the ion is spread evenly which is shown by using brackets
  - The charge on each ion is written at the top right-hand corner

Ionic bonds between Group I & Group VII elements

Group I & Group VII Ions

- Sodium is a Group 1 metal so **loses** one outer electron to another atom to gain a full outer shell of electrons
- A positive sodium ion with the charge 1+ is formed, Na^+
- Chlorine is a Group 7 non-metal so gains one electron to have a full outer shell of electrons
- A negative chloride ion with a charge of 1– is formed, Cl^-
- The ions are then attracted to one another and held together by **electrostatic forces**
- The formula of the ionic compound is thus NaCl
- The large square brackets should encompass each atom and the charge should be in superscript and on the right-hand side, outside the brackets

Diagram representing the formation of the ionic bond in sodium chloride

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